Lesson notes · for you

The sensors in your phone

IoT · Level 1 · Lesson 2

Time
45 min

Materials
A phone with phyphox, a compass app or a sound-meter app (with a parent's OK); Paper and a pencil for your table; No phone: squared paper and a pencil

Learning goals

- Name at least four sensors in a phone and what each is for.
- Explain that a camera is millions of light sensors that make a grid of numbers.
- Measure something with a phone sensor app.

Activity: Measure with your phone

1. Ask a parent or guardian first. Install phyphox (free) from the official app store, or use the compass app already on the phone.
2. Pick light, loudness or the compass direction. Light works in phyphox on most Android phones, not on iPhones.
3. Choose two places, like by a window and at the back of the room, or the classroom and the library. Hold the phone the same way each time.
4. Make a table: place, readings 1, 2 and 3, and the unit (lux for light, decibels for loudness, degrees for direction).
5. Which place had the bigger number, and why? Write two sentences. No phone: make a 10 × 10 number picture on squared paper instead.

Ask before installing anything, and use only the official app store. Light readings work on most Android phones, not on iPhones. No phone? The number picture on squared paper teaches the camera's idea just as well.

Discussion

- Which sensor would you miss most if your phone lost it?
- Why might two phones give different numbers in the same place?
- What could the sensors in your phone tell an app about your day?

Check yourself

- Name four phone sensors and one job for each.

Try it this week
With a parent's OK, measure one thing in two places with your phone, or make a number picture on squared paper.

Worksheet

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Name _____ Class _____ Date _____

1 **You take a photo of the school garden with a 12-megapixel camera. What does the phone actually store?**

- ◯ A tiny copy of the scene, like a printed photo inside the phone
- ◯ One number for the whole photo: how bright the garden was
- ◯ About 12 million pixels, each stored as three numbers

2 **Match what your phone does to the sensor behind it.**

The screen turns when you turn the phone	• • • •	GPS receiver
The screen goes dark at your ear in a call	• • • •	Compass sensor
The screen dims in a dark room	• • • •	Light sensor
Your map faces the way you're walking	• • • •	Proximity sensor
Your position shows on a saved map, with no data	• • • •	Accelerometer (motion sensor)

3 **Your phone's GPS needs mobile data or Wi-Fi to work out where you are.**

◯ True ◯ False

4 **Work out how many numbers a photo holds.**

A 12-megapixel photo has about _____ million pixels. With 3 numbers per pixel (red, green, blue), that's about _____ million numbers.

Word bank: 12 · 36 · 4 · 15

5 **Amani's screen stays on during calls, and her cheek keeps pressing buttons. What could be the cause?**

- ☐ The light sensor is broken
- ☐ A thick screen protector covers the proximity sensor
- ☐ The proximity sensor is dirty or broken
- ☐ The microphone is too quiet
- ☐ The GPS is switched off

Which phone sensor surprised you most? What could it tell an app about you if the app used it without asking?